Notes on Thermophoretic Force

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1 General formalism for rigid-body motion in a thermal gas cell

The disk and cylinder examples in Sections 2 and 3 exploit specific geometric symmetries to simplify the force expressions. In this section we derive the equations of motion for a rigid body of completely arbitrary shape, making no symmetry assumption. The only assumption throughout Sections 1.1–1.6 is that gravity acts along $-\hat{Z}$. The temperature gradient ∇T is treated as a general vector with arbitrary direction. Symmetric bodies such as cylinders and spheroids emerge as special cases of this formulation, which are discussed in Section 1.7.

1.1 Reference frames, rotation matrix, and inertia tensor

Lab frame. We work in a fixed inertial frame $(\hat{X}, \hat{Y}, \hat{Z})$. The only assumption placed on the external fields is that gravity acts along $-\hat{Z}$:

$$\boldsymbol{g} = -g\,\hat{Z}. \tag{1}$$

The macroscopic temperature gradient ∇T is a prescribed vector field in the lab frame. No assumption is made about its direction.

Body frame and inertia tensor. We attach a frame $(\hat{e}_1, \hat{e}_2, \hat{e}_3)$ to the rigid body with its origin at the centre of mass (CoM), aligned with the principal axes of inertia so that the inertia tensor is diagonal:

$$\mathbf{I} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}, \tag{2}$$

where I_1, I_2, I_3 are the three principal moments of inertia. In general all three are distinct, $I_1 \neq I_2 \neq I_3$.

Rotation matrix. The instantaneous orientation of the body is described by the rotation matrix $R \in \mathrm{SO}(3)$, defined so that a vector $\boldsymbol{u}$ with components in the body frame has

lab-frame components $\boldsymbol{u} = R\boldsymbol{u}^b$, with inverse $\boldsymbol{u}^b = R^\top \boldsymbol{u}$. In particular, the temperature gradient and the CoM velocity in the body frame are

$$(\nabla T)^b = R^\top \nabla T, \qquad \boldsymbol{v}^b = R^\top \boldsymbol{v}, \tag{3}$$

where $\boldsymbol{v} = \dot{\boldsymbol{x}}$ is the lab-frame CoM velocity. The evolution of R in time is governed by the quaternion kinematics derived in Section 1.5.

1.2 Gas-interaction tensors for an arbitrary shape

In the low-velocity, small-temperature-gradient limit all forces and torques respond linearly to their driving fields. We introduce four tensors in the natural order: the two forces (from temperature gradient and from velocity) followed by the two torques (from temperature gradient and from angular velocity).

Thermophoretic force tensor $\mathbf{f_{th}}$ (symmetric, 6 DOF). Thermophoresis drives particles from hot toward cold. The tensor $\mathbf{f}_{\text{th}}$ relates the temperature gradient to the thermophoretic force in the body frame:

$$\boldsymbol{F}^b_{\text{th}} = -\mathbf{f}_{\text{th}} \, (\nabla \ln T)^b, \qquad \mathbf{f}_{\text{th}} = \begin{pmatrix} f_{11} & f_{12} & f_{13} \\ f_{12} & f_{22} & f_{23} \\ f_{13} & f_{23} & f_{33} \end{pmatrix}. \tag{4}$$

$\mathbf{f}_{\text{th}}$ is symmetric positive-definite (6 DOF). The off-diagonal entry f_{ij} means a gradient along $\hat{e}_j$ drives a force component along $\hat{e}_i$.

Drag force tensor $\mathbf{f_{drag}}$ (symmetric, 6 DOF). The tensor $\mathbf{f}_{\text{drag}}$ relates the CoM velocity to the drag force in the body frame:

$$\boldsymbol{F}^b_{\text{drag}} = -\mathbf{f}_{\text{drag}} \, \boldsymbol{v}^b, \qquad \mathbf{f}_{\text{drag}} = \begin{pmatrix} d_{11} & d_{12} & d_{13} \\ d_{12} & d_{22} & d_{23} \\ d_{13} & d_{23} & d_{33} \end{pmatrix}. \tag{5}$$

$\mathbf{f}_{\text{drag}}$ is symmetric positive-definite (6 DOF): drag power $P = -\boldsymbol{v}^\top \mathbf{f}_{\text{drag}} \boldsymbol{v} < 0$ for all $\boldsymbol{v}$ forces positive-definiteness, and only the symmetric part contributes to P.

Thermophoretic torque tensor $\mathbf{T_{th}}$ (9 DOF). The temperature gradient can exert a net torque on a non-symmetric body:

$$\boldsymbol{\tau}^b_{\text{th}} = -\mathbf{T}_{\text{th}} \, (\nabla \ln T)^b, \qquad \mathbf{T}_{\text{th}} = \begin{pmatrix} t_{11} & t_{12} & t_{13} \\ t_{21} & t_{22} & t_{23} \\ t_{31} & t_{32} & t_{33} \end{pmatrix}. \tag{6}$$

$\mathbf{T}_{\text{th}}$ is in general a full 3×3 matrix (9 DOF) with no symmetry required. For bodies with three mutually perpendicular mirror planes $\mathbf{T}_{\text{th}} = 0$.

Rotational drag tensor $\mathbf{T}_{\text{drag}}$ (symmetric, 6 DOF). The tensor $\mathbf{T}_{\text{drag}}$ relates angular velocity to the resistive torque in the body frame:

$$\boldsymbol{\tau}^b_{\text{drag}} = -\mathbf{T}_{\text{drag}}\,\boldsymbol{\omega}^b, \qquad \mathbf{T}_{\text{drag}} = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{12} & r_{22} & r_{23} \\ r_{13} & r_{23} & r_{33} \end{pmatrix}. \tag{7}$$

$\mathbf{T}_{\text{drag}}$ is symmetric positive-definite (6 DOF) by the same dissipation argument as $\mathbf{f}_{\text{drag}}$.

1.3 Force and torque summary

The total force and torque on the body in the body frame are:

$$\boldsymbol{F}^b = \boldsymbol{F}^b_{\text{th}} + \boldsymbol{F}^b_{\text{drag}} = -\mathbf{f}_{\text{th}}\,(\nabla \ln T)^b - \mathbf{f}_{\text{drag}}\,\boldsymbol{v}^b, \tag{8}$$

$$\boldsymbol{\tau}^b = \boldsymbol{\tau}^b_{\text{th}} + \boldsymbol{\tau}^b_{\text{drag}} = -\mathbf{T}_{\text{th}}\,(\nabla \ln T)^b - \mathbf{T}_{\text{drag}}\,\boldsymbol{\omega}^b. \tag{9}$$

These four tensors and their driving fields can be collected into a single 6×6 linear response:

$$\begin{pmatrix} \boldsymbol{F}^b \\ \boldsymbol{\tau}^b \end{pmatrix} = -\begin{pmatrix} \mathbf{f}_{\text{th}} & \mathbf{f}_{\text{drag}} \\ \mathbf{T}_{\text{th}} & \mathbf{T}_{\text{drag}} \end{pmatrix} \begin{pmatrix} (\nabla \ln T)^b \\ \boldsymbol{v}^b \end{pmatrix}. \tag{10}$$

Note that the driving fields are $\nabla \ln T$ and $\boldsymbol{v}$ (not $\boldsymbol{\omega}$); the rotational responses τ_{drag} and the translational drag F_{drag} both arise from velocity fields in the gas. The four blocks have different physical dimensions:

$$[\mathbf{f}_{\text{th}}] = F{\cdot}L, \quad [\mathbf{f}_{\text{drag}}] = F{\cdot}T/L, \quad [\mathbf{T}_{\text{th}}] = F{\cdot}L^2, \quad [\mathbf{T}_{\text{drag}}] = F{\cdot}L^2{\cdot}T/L. \tag{11}$$

The total DOF for a body of arbitrary shape is $6 + 6 + 9 + 6 = 27$ (plus 6 for the inertia tensor). See Section 1.7 for how symmetry reduces this.

1.4 Equations of motion

Substituting the forces (8) and torques (9) into Newton's second law and Euler's equation, and expressing $(\nabla \ln T)^b = R^\top \nabla \ln T$, $\boldsymbol{v}^b = R^\top \boldsymbol{v}$, the complete equations of motion for the state $(\boldsymbol{x}, \boldsymbol{v}, R, \boldsymbol{\omega}^b)$ are:

Equations of motion

$$\dot{\boldsymbol{x}} = \boldsymbol{v}, \tag{12}$$

$$M\dot{\boldsymbol{v}} = -Mg\,\hat{Z} - R\big[\mathbf{f}_{\text{th}}\,R^\top \nabla \ln T + \mathbf{f}_{\text{drag}}\,R^\top \boldsymbol{v}\big], \tag{13}$$

$$\dot{R} = R\,\widehat{\boldsymbol{\omega}}^b, \tag{14}$$

$$\mathbf{I}\dot{\boldsymbol{\omega}}^b = -\boldsymbol{\omega}^b \times (\mathbf{I}\boldsymbol{\omega}^b) - \mathbf{T}_{\text{th}}\,R^\top \nabla \ln T - \mathbf{T}_{\text{drag}}\,\boldsymbol{\omega}^b. \tag{15}$$

Here $\widehat{\boldsymbol{a}}$ denotes the 3×3 skew-symmetric matrix associated with a vector $\boldsymbol{a}$, defined by $\widehat{\boldsymbol{a}}\boldsymbol{u} = \boldsymbol{a}\times\boldsymbol{u}$ for all $\boldsymbol{u}$, with entries $\widehat{\boldsymbol{a}}_{ij} = -\varepsilon_{ijk}a_k$. In (13) gravity acts downward ($-\hat{Z}$) and the temperature gradient ∇T is also taken along $-\hat{Z}$ (hot above cold), so the thermophoretic force $-\mathbf{f}_{\text{th}}R^\top\nabla\ln T$ points upward and can levitate the particle. Equation (14) is equivalent to the lab-frame form $\dot{R} = \widehat{\boldsymbol{\omega}}^{\text{lab}}R$ with $\boldsymbol{\omega}^{\text{lab}} = R\boldsymbol{\omega}^b$.

Gyroscopic term. The term $\boldsymbol{\omega}^b \times (\mathbf{I}\boldsymbol{\omega}^b)$ in Eq. (15) is the gyroscopic (Euler) term arising from the non-inertial body frame:

$$\boldsymbol{\omega}^b \times (\mathbf{I}\boldsymbol{\omega}^b) = \begin{pmatrix} (I_3 - I_2)\omega_2^b \omega_3^b \\ (I_1 - I_3)\omega_3^b \omega_1^b \\ (I_2 - I_1)\omega_1^b \omega_2^b \end{pmatrix}. \tag{16}$$

It vanishes for a sphere ($I_1 = I_2 = I_3$) and when $\boldsymbol{\omega}^b$ is parallel to a principal axis.

Practical implementation. For numerical integration, $R \in \mathrm{SO}(3)$ is represented by a unit quaternion $\boldsymbol{q}$ to avoid the gimbal-lock singularity of Euler angles. Equation (14) then becomes $\dot{\boldsymbol{q}} = \frac{1}{2} Q(\boldsymbol{q})\boldsymbol{\omega}^b$ (see Section 1.5).

1.5 Attitude kinematics — quaternion representation

Euler angles develop a coordinate singularity (gimbal lock) at pitch $\pm 90^\circ$. We instead represent the orientation by a unit quaternion

$$\boldsymbol{q} = (q_0, q_1, q_2, q_3) \in \mathbb{R}^4, \quad |\boldsymbol{q}|^2 \equiv q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1. \tag{17}$$

The scalar part is $q_0 = \cos(\psi/2)$ and the vector part is $\vec{q} = (q_1, q_2, q_3) = \hat{k}\sin(\psi/2)$, where $\hat{k}$ is the unit rotation axis and ψ is the rotation angle.

Rotation matrix from quaternion.

$$R(\boldsymbol{q}) = (q_0^2 - |\vec{q}|^2)\mathbf{I}_3 + 2\vec{q}\,\vec{q}^{\,\top} + 2q_0\,\widehat{\vec{q}}, \qquad \widehat{\vec{q}} = \begin{pmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{pmatrix}. \tag{18}$$

Explicit axis-angle form.

$$R(\hat{k}, \psi) = \cos\psi\, \mathbf{I}_3 + (1 - \cos\psi)\hat{k}\hat{k}^\top + \sin\psi\, \widehat{\hat{k}}, \tag{19}$$

with $c \equiv \cos\psi$, $s \equiv \sin\psi$:

$$R(\hat{k}, \psi) = \begin{pmatrix} k_1^2(1-c) + c & k_1k_2(1-c) - k_3 s & k_1k_3(1-c) + k_2 s \\ k_1k_2(1-c) + k_3 s & k_2^2(1-c) + c & k_2k_3(1-c) - k_1 s \\ k_1k_3(1-c) - k_2 s & k_2k_3(1-c) + k_1 s & k_3^2(1-c) + c \end{pmatrix}. \tag{20}$$

Kinematic differential equation.

$$\dot{\boldsymbol{q}} = \tfrac{1}{2} Q(\boldsymbol{q})\boldsymbol{\omega}^b, \qquad Q(\boldsymbol{q}) = \begin{pmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{pmatrix}. \tag{21}$$

To prevent numerical drift, $\boldsymbol{q}$ must be renormalised after each integration step: $\boldsymbol{q} \leftarrow \boldsymbol{q}/|\boldsymbol{q}|$.

1.6 State-vector formulation

The complete dynamical state of the rigid body is encoded in 13 scalar variables:

$$\boldsymbol{s} = \big(\underbrace{\boldsymbol{x}}_{3}, \underbrace{\boldsymbol{v}}_{3}, \underbrace{\boldsymbol{q}}_{4}, \underbrace{\boldsymbol{\omega}^b}_{3}\big) \in \mathbb{R}^{13}. \tag{22}$$

Complete equations of motion (quaternion form)

State vector: $\boldsymbol{s} = (\boldsymbol{x}, \boldsymbol{v}, \boldsymbol{q}, \boldsymbol{\omega}^b) \in \mathbb{R}^{13}$

Orientation: $R(\boldsymbol{q}) = (q_0^2 - |\vec{q}|^2)\mathbf{I}_3 + 2\vec{q}\,\vec{q}^{\top} + 2q_0\,\widehat{\vec{q}}, \quad \widehat{\vec{q}} = \begin{pmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{pmatrix}$

Equations of motion:

$$\dot{\boldsymbol{x}} = \boldsymbol{v}, \tag{23}$$

$$M\dot{\boldsymbol{v}} = -Mg\,\hat{Z} - R(\boldsymbol{q})\big[\mathbf{f}_{\text{th}}\,R(\boldsymbol{q})^{\top}\nabla \ln T + \mathbf{f}_{\text{drag}}\,R(\boldsymbol{q})^{\top}\boldsymbol{v}\big], \tag{24}$$

$$\dot{\boldsymbol{q}} = \tfrac{1}{2}Q(\boldsymbol{q})\boldsymbol{\omega}^b, \tag{25}$$

$$\mathbf{I}\dot{\boldsymbol{\omega}}^b = -\boldsymbol{\omega}^b \times (\mathbf{I}\boldsymbol{\omega}^b) - \mathbf{T}_{\text{th}}\,R(\boldsymbol{q})^{\top}\nabla \ln T - \mathbf{T}_{\text{drag}}\,\boldsymbol{\omega}^b. \tag{26}$$

Renormalise after each step: $\boldsymbol{q} \leftarrow \boldsymbol{q}/|\boldsymbol{q}|$

1.7 Tensor simplification by geometric symmetry

The 27 DOF of a generic body reduce dramatically when the body has geometric symmetry. We organise the simplification into three levels, from most general to most symmetric.

Level 1 — Generic asymmetric body (27 DOF). All four tensors are full matrices with their maximum DOF:

$$\mathbf{f}_{\text{th}} : 6, \quad \mathbf{f}_{\text{drag}} : 6, \quad \mathbf{T}_{\text{th}} : 9, \quad \mathbf{T}_{\text{drag}} : 6. \tag{27}$$

Examples: a helix, an asymmetric two-sphere aggregate, any body lacking mirror-plane symmetry. The thermophoretic torque $\mathbf{T}_{\text{th}} \neq 0$ drives steady rotation and, via the coupling to translation, circular orbits (Section 2.2).

Level 2 — Three mutually perpendicular mirror planes (18 DOF). The mirror symmetry cancels all off-diagonal elements of $\mathbf{f}_{\text{th}}$, $\mathbf{f}_{\text{drag}}$, $\mathbf{T}_{\text{drag}}$, and forces $\mathbf{T}_{\text{th}} = 0$ (Garcia-Ybarra & Rosner 1989):

$$\mathbf{f}_{\text{th}} = \text{diag}(f_1, f_2, f_3), \quad \mathbf{f}_{\text{drag}} = \text{diag}(d_1, d_2, d_3), \quad \mathbf{T}_{\text{drag}} = \text{diag}(r_1, r_2, r_3), \quad \mathbf{T}_{\text{th}} = 0. \tag{28}$$

DOF: $3 + 3 + 0 + 3 = 9$. Bodies in this class: spheroids, cylinders, disks, rectangular blocks. With $\mathbf{T}_{\text{th}} = 0$ there is no thermophoretic torque; the particle drifts without spinning.

Level 3a — Uniaxial symmetry, $C_{\infty v}$ (6 DOF). With a single symmetry axis $\hat{e}_3$, the two transverse directions are equivalent: $f_1 = f_2 \equiv f_\perp$, $f_3 \equiv f_\parallel$ (and similarly for $\mathbf{f}_{\text{drag}}$, $\mathbf{T}_{\text{drag}}$), $\mathbf{T}_{\text{th}} = 0$:

$$\mathbf{f}_{\text{th}} = \text{diag}(f_\perp, f_\perp, f_\parallel), \quad \mathbf{f}_{\text{drag}} = \text{diag}(d_\perp, d_\perp, d_\parallel), \quad \mathbf{T}_{\text{drag}} = \text{diag}(r_\perp, r_\perp, r_\parallel). \tag{29}$$

DOF: $2 + 2 + 0 + 2 = 6$. Bodies: cylinders, spheroids.

Level 3b — Sphere, SO(3) symmetry (3 DOF). Full isotropy collapses every tensor to a scalar multiple of $\mathbf{I}_3$:

$$\mathbf{f}_{\text{th}} = f_0\mathbf{I}_3, \quad \mathbf{f}_{\text{drag}} = d_0\mathbf{I}_3, \quad \mathbf{T}_{\text{drag}} = r_0\mathbf{I}_3, \quad \mathbf{T}_{\text{th}} = 0, \quad I_1 = I_2 = I_3. \tag{30}$$

DOF: $1 + 1 + 0 + 1 = 3$. The gyroscopic term vanishes, the rotational equation decouples from translation, and the thermophoretic force reduces to $\boldsymbol{F}_{\text{th}} = -f_0 \nabla \ln T$, consistent with Waldmann's free-molecule result.

Table 1: Degrees of freedom by symmetry class (excluding the inertia tensor). $\mathbf{T}_{\text{th}}$ counts 9 DOF when non-zero since no symmetry argument forces it to be symmetric.

Body	$\mathbf{f}_{\text{th}}$	$\mathbf{f}_{\text{drag}}$	$\mathbf{T}_{\text{th}}$	$\mathbf{T}_{\text{drag}}$	Total
Generic asymmetric	6	6	9	6	27
Three mirror planes	3	3	0	3	9
Uniaxial ($C_{\infty v}$)	2	2	0	2	6
Sphere (SO(3))	1	1	0	1	3

2 Near-sphere perturbation theory

The general equations of motion in Section 1 contain five tensors with up to 24 independent parameters. For a particle that is nearly spherical, the deviation from sphere symmetry is small and the tensors can be expanded perturbatively. This approach gives analytical insight into how shape asymmetry couples translation to rotation, and predicts a distinctive helical trajectory that is absent for a perfect sphere.

2.1 Perturbative expansion of the tensors

Let $\varepsilon \ll 1$ be a dimensionless measure of the deviation from spherical symmetry (e.g. the fractional difference between the longest and shortest semi-axes). We expand each tensor around its spherical value.

Thermophoretic force tensor. For a sphere $\mathbf{f}_{\text{th}} = \gamma_0\mathbf{I}_3$ with $\gamma_0 = \eta(\kappa/\bar{v})\ell^2$. A near-sphere breaks the three-fold degeneracy and, for a generic asymmetric shape, also introduces

off-diagonal couplings. The full $O(\varepsilon)$ expansion is therefore a complete symmetric 3×3 matrix:

$$\mathbf{f}_{\text{th}} = \gamma_0 \mathbf{I}_3 + \varepsilon\, \delta\mathbf{f}_{\text{th}}, \qquad \delta\mathbf{f}_{\text{th}} = \gamma_0 \begin{pmatrix} g_{11} & g_{12} & g_{13} \\ g_{12} & g_{22} & g_{23} \\ g_{13} & g_{23} & g_{33} \end{pmatrix}, \quad g_{11} + g_{22} + g_{33} = 0, \tag{31}$$

where $g_{ij} = O(1)$. Off-diagonal entries g_{ij} $(i \neq j)$ mean that a temperature gradient along $\hat{e}_j$ produces a thermophoretic force component along $\hat{e}_i$. Tracelessness follows because the sphere value γ_0 accounts for the mean response.

Resistance and rotational drag tensors. By the same argument, $\mathbf{f}_{\text{drag}}$ and $\mathbf{T}_{\text{drag}}$ each acquire a full symmetric $O(\varepsilon)$ correction:

$$\mathbf{f}_{\text{drag}} = \xi_0 \mathbf{I}_3 + \varepsilon\, \delta\mathbf{f}_{\text{drag}}, \qquad \mathbf{T}_{\text{drag}} = c_0 \mathbf{I}_3 + \varepsilon\, \delta\mathbf{T}_{\text{drag}}, \tag{32}$$

where $\delta\mathbf{f}_{\text{drag}}$ and $\delta\mathbf{T}_{\text{drag}}$ are symmetric traceless 3×3 matrices.

Thermophoretic torque tensor. For a perfect sphere $\mathbf{T}_{\text{th}} = 0$. For a near-sphere the broken symmetry produces a non-zero thermophoretic torque at $O(\varepsilon)$:

$$\mathbf{T}_{\text{th}} = \varepsilon\, \delta\mathbf{T}_{\text{th}}, \qquad \delta\mathbf{T}_{\text{th}} = O(1) \text{ (general } 3 \times 3 \text{ matrix)}. \tag{33}$$

Both diagonal and off-diagonal entries are generically non-zero.

Note on gyrophoretic coupling. Since no experimental evidence currently constrains the relationship and $\mathbf{T}_{\text{th}}$; any gyrophoretic coupling is $O(\varepsilon^2)$ and is neglected.

Summary of orders (to leading order in ε). Collecting Eqs. (31)–(33):

$$\boxed{\mathbf{f}_{\text{th}} = f_0 \mathbf{I}_3 + \varepsilon\, \delta\mathbf{f}_{\text{th}}, \quad \mathbf{f}_{\text{drag}} = d_0 \mathbf{I}_3 + \varepsilon\, \delta\mathbf{f}_{\text{drag}}, \quad \mathbf{T}_{\text{th}} = \varepsilon\, \delta\mathbf{T}_{\text{th}}, \quad \mathbf{T}_{\text{drag}} = r_0 \mathbf{I}_3 + \varepsilon\, \delta\mathbf{T}_{\text{drag}}.} \tag{34}$$

Here $f_0 \equiv \gamma_0$, $d_0 \equiv \xi_0$, $r_0 \equiv c_0$ are the scalar sphere values, and the four correction tensors $\delta\mathbf{f}_{\text{th}}$, $\delta\mathbf{f}_{\text{drag}}$, $\delta\mathbf{T}_{\text{th}}$, $\delta\mathbf{T}_{\text{drag}}$ are defined in Eqs. (31)–(33) above. At $O(\varepsilon^0)$ each tensor is isotropic; all anisotropy and the thermophoretic torque enter at $O(\varepsilon)$ through these corrections.

2.2 Rotational dynamics and orientation selection

Equations of motion for $(\boldsymbol{\omega}^b, R)$. For a near-sphere ($\mathbf{I} = I_0 \mathbf{I}_3 + O(\varepsilon)$) the Euler equation (15) in the body frame, keeping only $O(\varepsilon)$ terms, is:

$$I_0 \dot{\boldsymbol{\omega}}^b = G\varepsilon\, \delta\mathbf{T}_{\text{th}}\, R^\top \hat{Z} - c_0 \boldsymbol{\omega}^b + O(\varepsilon^2), \tag{35}$$

$$\dot{R} = R\, \widehat{\boldsymbol{\omega}}^b, \tag{36}$$

where we substituted $(\nabla \ln T)^b = R^\top \nabla \ln T = -G R^\top \hat{Z}$ (with $G = |\nabla \ln T| = |\nabla T|/T > 0$ and $\nabla \ln T = -G\hat{Z}$ pointing downward), so that $-\delta\mathbf{T}_{\text{th}} (\nabla \ln T)^b = +G\, \delta\mathbf{T}_{\text{th}} R^\top \hat{Z}$, and dropped the gyroscopic term $\boldsymbol{\omega}^b \times (I_0 \boldsymbol{\omega}^b)$ (it is $O(\varepsilon^2)$ since $\boldsymbol{\omega}^b = O(\varepsilon)$). Equations (35)–(36) together form a closed system for $(\boldsymbol{\omega}^b, R) \in \mathbb{R}^3 \times \mathrm{SO}(3)$.

Reduced form via $\boldsymbol{u} = R^\top \hat{Z}$. It is convenient to project onto the body-frame gradient direction $\boldsymbol{u} \equiv R^\top \hat{Z}$, $|\boldsymbol{u}| = 1$. Differentiating Eq. (36): $\dot{\boldsymbol{u}} = \dot{R}^\top \hat{Z} = -\widehat{\boldsymbol{\omega}}^b \boldsymbol{u} = -\boldsymbol{\omega}^b \times \boldsymbol{u}$. Equations (35)–(36) reduce to:

$$I_0 \dot{\boldsymbol{\omega}}^b = G\varepsilon\, \delta\mathbf{T}_{\rm th} \boldsymbol{u} - c_0 \boldsymbol{\omega}^b, \qquad \dot{\boldsymbol{u}} = -\boldsymbol{\omega}^b \times \boldsymbol{u}, \tag{37}$$

a closed 6-DOF ODE on $\mathbb{R}^3 \times S^2$, exact to $O(\varepsilon)$. Once $\boldsymbol{u}(t)$ is known, $R(t)$ can be recovered by integrating Eq. (36).

Steady states: eigenvalue problem. Setting $\dot{\boldsymbol{\omega}}^b = 0$ and $\dot{\boldsymbol{u}} = 0$ in Eq. (37):

$$c_0 \boldsymbol{\omega}^b = G\varepsilon\, \delta\mathbf{T}_{\rm th} \boldsymbol{u}, \qquad \boldsymbol{\omega}^b \times \boldsymbol{u} = 0. \tag{38}$$

The second condition requires $\boldsymbol{\omega}^b \parallel \boldsymbol{u}$. Substituting into the first:

$$\delta\mathbf{T}_{\rm th} \boldsymbol{u} = \lambda \boldsymbol{u}, \qquad \boldsymbol{\omega}^b = \frac{G\varepsilon\lambda}{c_0} \boldsymbol{u}. \tag{39}$$

The steady-state body-frame gradient $\boldsymbol{u}$ must be an eigenvector of $\delta\mathbf{T}_{\rm th}$ with eigenvalue λ. For a general 3×3 matrix $\delta\mathbf{T}_{\rm th}$ there are three such eigenmotions.

The three eigenmotions in the lab frame. Let $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$ be the eigenvectors of $\delta\mathbf{T}_{\rm th}$ with real eigenvalues $\lambda_1 \le \lambda_2 \le \lambda_3$ (real eigenvalues exist when $\delta\mathbf{T}_{\rm th}$ is symmetric; for a non-symmetric matrix the analysis extends to the real Schur form). Each eigenvector defines a distinct steady state. At the fixed point $\boldsymbol{u} = \boldsymbol{e}_k$ we have $R^\top \hat{Z} = \boldsymbol{e}_k$, equivalently $R\boldsymbol{e}_k = \hat{Z}$: the body orients so that its $\boldsymbol{e}_k$-axis aligns with the vertical (the temperature-gradient direction).

From Eq. (39) the angular velocity in the body frame is $\boldsymbol{\omega}^b_k = (G\varepsilon\lambda_k/c_0)\boldsymbol{e}_k$, a spin about the body axis $\boldsymbol{e}_k$. Transforming to the lab frame gives

$$\boldsymbol{\Omega}_k = R\boldsymbol{\omega}^b_k = \frac{G\varepsilon\lambda_k}{c_0}\, R\boldsymbol{e}_k = \frac{G\varepsilon\lambda_k}{c_0}\, \hat{Z}. \tag{40}$$

The lab-frame angular velocity is exactly along $\hat{Z}$ for all three eigenmotions — this is a direct consequence of the fixed-point geometry $R\boldsymbol{e}_k = \hat{Z}$. Physically, the body spins about the temperature-gradient direction.

Stability of the three eigenmotions. Linearising Eq. (37) about the fixed point $\boldsymbol{u} = \boldsymbol{e}_k$ with perturbation $\delta\boldsymbol{u} = \sum_{j\neq k} \delta u_j \boldsymbol{e}_j$, the deviation decays or grows at rate:

$$\sigma_{kj} = \frac{G\varepsilon(\lambda_k - \lambda_j)\lambda_j}{I_0 c_0} \quad (j \neq k). \tag{41}$$

The fixed point $\boldsymbol{e}_k$ is stable if $\sigma_{kj} < 0$ for all $j \neq k$. For λ_k the largest eigenvalue, $(\lambda_k - \lambda_j) > 0$ and $\lambda_j < \lambda_k$, so $\sigma_{kj} < 0$ requires $\lambda_j < 0$; alternatively, if all eigenvalues are positive, stability requires $\lambda_k < \lambda_j$. In general the sign of λ depends on the body: the stable attractor is the orientation $\boldsymbol{e}_k$ for which $\sigma_{kj} < 0$ for all $j \neq k$, i.e. $(\lambda_k - \lambda_j)\lambda_j < 0$ for all j, which is guaranteed

when λ_k is the eigenvalue of largest magnitude with sign opposite to all others, or when all λ_j have the same sign and λ_k is the largest. For the common case $\lambda_3 > \lambda_2 > \lambda_1 > 0$ the largest eigenvalue λ_3 gives $\sigma_{3j} = G\varepsilon(\lambda_3 - \lambda_j)\lambda_j/(I_0 c_0) > 0$ (all terms positive), which is *unstable*. The stable attractor is then $\boldsymbol{e}_1$ (smallest eigenvalue). The general rule is: *the stable orientation is the eigenvector of* $\delta\mathbf{T}_{th}$ *that maximises* $V(\boldsymbol{u}) = -\boldsymbol{u}^\top \delta\mathbf{T}_{th}\boldsymbol{u}$, i.e. the eigenvector with the most-negative eigenvalue when the gradient drives alignment, or equivalently the orientation that minimises the thermophoretic torque-gradient product. The three eigenmotions therefore have the following stability character:

Eigenmotion	Eigenvalue	Stability
$\boldsymbol{e}_1$	λ_1 (smallest)	unstable (repeller)
$\boldsymbol{e}_2$	λ_2 (middle)	saddle (unstable)
$\boldsymbol{e}_3$	λ_3 (largest)	**stable attractor**

All generic initial conditions converge to eigenmotion 3: the body aligns the axis with the largest eigenvalue of $\delta\mathbf{T}_{\text{th}}$ with the temperature gradient. This maximises the Lyapunov function $V(\boldsymbol{u}) = \boldsymbol{u}^\top \delta\mathbf{T}_{\text{th}}\boldsymbol{u}$, which measures the projection of the thermophoretic torque onto the gradient direction.

Physical interpretation. The body rotates until the torque $\delta\mathbf{T}_{\text{th}}\boldsymbol{u}$ is parallel to the driving field $\boldsymbol{u}$, leaving no torque component perpendicular to $\boldsymbol{u}$ to drive further reorientation. Among the three possible aligned orientations only the one with the largest eigenvalue is stable: a perturbation away from this orientation generates a restoring torque that drives the body back. This is not a minimisation of rotational energy $E_{\text{rot}} \propto \lambda_k^2$ (which is also maximised at λ_3) but rather a maximisation of the torque-gradient alignment V.

2.3 Translational dynamics and general steady motion

Once the angular velocity $\boldsymbol{\omega}^b$ is determined from the rotational steady state (Section 2.2), the translational equation (13) in the body frame at steady state ($\dot{\boldsymbol{v}} = 0$) becomes, using $\nabla \ln T = -G\hat{Z}$ and $\boldsymbol{u} = R^\top \hat{Z}$:

$$\mathbf{f}_{\text{drag}}\boldsymbol{v}^b = G\mathbf{f}_{\text{th}}\boldsymbol{u} - Mg\boldsymbol{u}. \tag{42}$$

Expanding with Eq. (34), $\mathbf{f}_{\text{th}} = \gamma_0\mathbf{I}_3 + \varepsilon\,\delta\mathbf{f}_{\text{th}}$ and $\mathbf{f}_{\text{drag}} = \xi_0\mathbf{I}_3 + \varepsilon\,\delta\mathbf{f}_{\text{drag}}$:

$$\xi_0\boldsymbol{v}^b = (\gamma_0 G - Mg)\boldsymbol{u} + \varepsilon\left[G\,\delta\mathbf{f}_{\text{th}} - \frac{\gamma_0 G - Mg}{\xi_0}\delta\mathbf{f}_{\text{drag}}\right]\boldsymbol{u}. \tag{43}$$

Levitation condition to first order in ε. The lab-frame vertical velocity is obtained by projecting Eq. (43) onto $\boldsymbol{u} = R^\top\hat{Z}$ (using $v_Z = \hat{Z}\cdot R\boldsymbol{v}^b = \boldsymbol{u}\cdot\boldsymbol{v}^b$):

$$v_Z = \frac{\gamma_0 G - Mg}{\xi_0} + \varepsilon G\,(\delta\mathbf{A})_{33}, \tag{44}$$

where we define the anisotropic mobility tensor

$$\delta\mathbf{A} \equiv \frac{1}{\xi_0}(\delta\mathbf{f}_{\text{th}} - A_0\,\delta\mathbf{f}_{\text{drag}})\,, \quad A_0 \equiv \frac{\gamma_0}{\xi_0}, \tag{45}$$

and $(\delta\mathbf{A})_{33} = \boldsymbol{e}_3 \cdot \delta\mathbf{A}\boldsymbol{e}_3$ is its component along the precession axis. The *full* levitation condition $\langle v_Z \rangle = 0$ applied to Eq. (44) gives a single equation for the required temperature-gradient magnitude:

$$G = \frac{Mg}{\gamma_0 + \varepsilon\,\xi_0\,(\delta\mathbf{A})_{33}} = \frac{Mg}{\gamma_0}\big[1 - \varepsilon A_0\,(\delta\mathbf{A})_{33} + O(\varepsilon^2)\big]\,. \tag{46}$$

The zeroth-order levitation gradient is $G_0 = Mg/\gamma_0$, and the $O(\varepsilon)$ correction shifts G by an amount proportional to $(\delta\mathbf{A})_{33}$: the asymmetry of the body modifies the gradient needed to keep the particle suspended. There is no separate constraint on $(\delta\mathbf{A})_{33}$; it is a property of the particle shape that the experimenter accommodates by tuning G.

First-order body-frame velocity. With the levitation gradient (46) substituted back into Eq. (43), the body-frame velocity becomes, to $O(\varepsilon)$:

$$\boldsymbol{v}^b = \varepsilon G_0\,\delta\mathbf{A}\boldsymbol{u} - \varepsilon G_0 A_0\,(\delta\mathbf{A})_{33}\,\boldsymbol{u} + O(\varepsilon^2). \tag{47}$$

The two terms together ensure $\boldsymbol{u}\cdot\boldsymbol{v}^b = 0$ to $O(\varepsilon)$: the body-frame velocity has *no projection along* $\boldsymbol{u}$, so there is no residual vertical drift. The lab-frame velocity is $\boldsymbol{v} = R\boldsymbol{v}^b$ and lies entirely in the plane perpendicular to $\hat{Z}$.

General steady solutions. We now enumerate the possible steady motions depending on the orientation state.

Case 1: Symmetric body ($\delta\mathbf{T}_{th} = 0$). Then $\boldsymbol{\omega}^b = 0$, R is constant, and Eq. (47) gives a constant body-frame velocity. The lab-frame velocity $\boldsymbol{v} = \varepsilon G\,R\,\delta\mathbf{A}\,R^\top\hat{Z}$ is also constant. This is straight-line drift in a fixed direction. At levitation, the drift is purely horizontal.

Case 2: Asymmetric body, transient phase. Before the orientation has converged to a fixed point, $R(t)$ evolves on the alignment timescale $\sim I_0/c_0$. The CoM trajectory is non-periodic and traces a curved path depending on the initial orientation.

Case 3: Asymmetric body, steady precession (attractor). At the fixed point $\boldsymbol{u} = \boldsymbol{e}_3$ the lab-frame angular velocity is $\boldsymbol{\Omega}_3 = (G\varepsilon\lambda_3/c_0)\hat{Z}$ (Eq. (40)), parallel to $\hat{Z}$. The body therefore precesses about $\hat{Z}$: $R(t) = R_Z(\Omega_3 t)R_0$ with $R_0\boldsymbol{e}_3 = \hat{Z}$. With the levitation gradient (46) imposed, $\boldsymbol{v}^b$ has no projection along $\boldsymbol{e}_3$, so the lab-frame velocity is entirely horizontal:

$$\boldsymbol{v}(t) = \varepsilon G_0\,R_Z(\Omega_3 t)\,\Pi_\perp\delta\mathbf{A}\,\boldsymbol{e}_3, \quad \Pi_\perp = \mathbf{I}_3 - \hat{Z}\hat{Z}^\top, \tag{48}$$

with $v_Z = 0$ by construction and $\boldsymbol{v}_\perp(t)$ having constant magnitude. Since $\boldsymbol{v}_\perp$ lies in the horizontal plane and $\boldsymbol{\Omega}_3 \parallel \hat{Z}$, we have $\boldsymbol{v}_\perp \perp \boldsymbol{\Omega}_3$. The two conditions for circular motion are therefore:

1. $\boldsymbol{\Omega}_3 \parallel \hat{Z}$ — from the fixed-point geometry $R\boldsymbol{e}_3 = \hat{Z}$.
2. $\boldsymbol{v}_\perp \perp \boldsymbol{\Omega}_3$ — automatically, since $\boldsymbol{v}_\perp$ lies in the horizontal plane.

This is circular orbit motion.

Circular orbit radius. For the levitated case (gradient G given by Eq. (46)), both the orbital speed $|\boldsymbol{v}_\perp| = \varepsilon G_0 |\Pi_\perp R_0\, \delta\mathbf{A}\, \boldsymbol{e}_3|$ and the spin rate $|\Omega_3| = G_0 \varepsilon |\lambda_3|/c_0$ are linear in εG_0, so their ratio is $O(1)$:

$$\Omega_3 = \frac{G_0 \varepsilon \lambda_3}{c_0}, \qquad R_{\text{orbit}} = \frac{|\boldsymbol{v}_\perp|}{|\Omega_3|} = \frac{c_0\, |\Pi_\perp R_0\, \delta\mathbf{A}\, \boldsymbol{e}_3|}{|\lambda_3|}. \tag{49}$$

Key features:

1. R_{orbit} is $O(1)$: it is independent of both ε and G_0, since the εG_0 factors cancel in the ratio $|\boldsymbol{v}_\perp|/|\Omega_3|$. R_{orbit} depends only on the particle shape through $\delta\mathbf{A}$ and λ_3.

2. $\Omega_3 \propto \varepsilon G_0$: a more asymmetric body or a stronger gradient spins faster, and the orbital speed $|\boldsymbol{v}_\perp| \propto \varepsilon G_0$ grows at the same rate, keeping the radius fixed.

3. If $\Pi_\perp R_0\, \delta\mathbf{A}\, \boldsymbol{e}_3 = 0$ (the anisotropic drift of eigenmotion 3 is purely along $\hat{Z}$), then $R_{\text{orbit}} = 0$: the particle hovers at a fixed point with no horizontal motion. This occurs when $\delta\mathbf{f}_{\text{th}}$ and $\delta\mathbf{f}_{\text{drag}}$ are simultaneously diagonal in the same basis as $\delta\mathbf{T}_{\text{th}}$; in this case the levitation correction in Eq. (46) accounts for all anisotropy and no horizontal orbit appears.

4. No orbit for a symmetric body: $\delta\mathbf{T}_{\text{th}} = 0$ gives $\lambda_3 = 0$, $\Omega_3 = 0$, and the particle drifts in a fixed direction (or hovers if the drift is vertical).

Summary of steady motions.

Condition	$\boldsymbol{\Omega}$	Trajectory
$\delta\mathbf{T}_{\text{th}} = 0$ (symmetric)	0	straight-line drift (or hover)
$\delta\mathbf{T}_{\text{th}} \neq 0$, transient	time-varying	curved, non-periodic
$\delta\mathbf{T}_{\text{th}} \neq 0$, eigenmotion 3	$\Omega_3 \hat{Z}$	horizontal circle (or hover)
Eigenmotion 1 or 2	$\Omega_{1,2} \hat{Z}$	circle (unstable)

2.4 Electromagnetic analogy and ponderomotive picture

The circular orbit derived in Section 2.3 has a striking structural resemblance to the cyclotron motion of a charged particle in crossed electric and magnetic fields. In this section we make that analogy precise, identify the effective electromagnetic quantities, and explain through the ponderomotive picture why the orbit radius is independent of the driving field strength.

The temperature field as vector potential. In electromagnetism a single vector potential $\boldsymbol{A}$ generates both the electric and magnetic fields through differential operations:

$$\boldsymbol{E} = -\partial_t \boldsymbol{A}, \qquad \boldsymbol{B} = \nabla \times \boldsymbol{A}. \tag{50}$$

The electric field arises from a *time* derivative of $\boldsymbol{A}$; the magnetic field from a *spatial* (curl) derivative. Both fields are linear in $\boldsymbol{A}$, so any ratio of their effects on the particle is independent of the overall amplitude of $\boldsymbol{A}$.

We propose the following identification. The logarithmic temperature gradient

$$\boldsymbol{\mathcal{A}} \equiv \nabla \ln T = \frac{\nabla T}{T} \tag{51}$$

plays the role of the vector potential $\boldsymbol{A}$. It is the single field from which all thermophoretic responses derive: Waldmann's free-molecule force is $\boldsymbol{F}_T \propto \nabla \ln T$, and every tensor in Sections 1–2 is contracted with $\nabla \ln T$. In the levitated steady state $\boldsymbol{\mathcal{A}} = -G_0 \hat{Z}$ is uniform and static, so the analogy with a propagating electromagnetic wave (where $\boldsymbol{A}$ oscillates in both space and time) is not exact; we return to this point below.

The two tensors as differential operators. The shape tensors $\mathbf{f}_{\rm th}$ and $\mathbf{T}_{\rm th}$ act on $\boldsymbol{\mathcal{A}}$ to produce the force and torque, respectively:

$$\boldsymbol{F}_{\rm th} = -\mathbf{f}_{\rm th}\,\boldsymbol{\mathcal{A}}, \qquad \boldsymbol{\tau}_{\rm th} = -\mathbf{T}_{\rm th}\,\boldsymbol{\mathcal{A}}. \tag{52}$$

Comparing with Eq. (50), the symmetric tensor $\mathbf{f}_{\rm th}$ plays the role of the *time-derivative operator* ∂_t: it extracts from the potential a conservative translational force, just as $-\partial_t \boldsymbol{A}$ produces the irrotational electric field that does work on the charge. The asymmetric tensor $\mathbf{T}_{\rm th}$ plays the role of the *curl operator* $\nabla\times$: it extracts a rotational response from the same potential, just as $\nabla \times \boldsymbol{A}$ produces the solenoidal magnetic field that drives precession without doing work.

The near-sphere perturbation theory of Section 2.1 makes this decomposition explicit. To leading order $\mathbf{f}_{\rm th} = f_0 \mathbf{I}_3 + \varepsilon\,\delta\mathbf{f}_{\rm th}$ is symmetric (the "electric" part) and $\mathbf{T}_{\rm th} = \varepsilon\,\delta\mathbf{T}_{\rm th}$ is generically asymmetric (the "magnetic" part), with both vanishing for a sphere ($\varepsilon = 0$): the sphere is the neutral, non-orbiting particle of this analogy. Shape asymmetry simultaneously switches on both an effective electric field (anisotropic translational drift) and an effective magnetic field (thermophoretic torque driving spin and orbital precession), from the same single source $\boldsymbol{\mathcal{A}}$.

Effective electric and magnetic fields. We define the effective fields seen by the particle as

$$\boldsymbol{E}_{\rm eff} \equiv -\varepsilon\,\delta\mathbf{f}_{\rm th}\,\boldsymbol{\mathcal{A}}, \qquad \boldsymbol{B}_{\rm eff} \equiv \varepsilon\,\delta\mathbf{T}_{\rm th}\,\boldsymbol{\mathcal{A}}, \tag{53}$$

so that both are $O(\varepsilon G_0)$: they vanish for a sphere and grow with asymmetry and gradient strength in the same proportion. The translational steady-state drift (Eq. (47)) is driven by $\boldsymbol{E}_{\rm eff}$, and the rotational steady state (Eq. (39)) is driven by $\boldsymbol{B}_{\rm eff}$ — precisely the roles of $\boldsymbol{E}$ and $\boldsymbol{B}$ in the Lorentz force.

The ponderomotive picture and amplitude-independence of $R_{\rm orbit}$. In electromagnetism the most natural situation in which *both* $\boldsymbol{E}$ and $\boldsymbol{B}$ derive from a single potential of amplitude $\mathcal{A}$ is a propagating electromagnetic wave, $\boldsymbol{A} \propto \mathcal{A}\cos(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)$, with $|\boldsymbol{E}| = \omega\mathcal{A}$ and $|\boldsymbol{B}| = k\mathcal{A}$. A charged particle in such a wave undergoes fast quiver oscillations at frequency ω superimposed on a slow drift. The slow drift is governed by the *ponderomotive force*

$$\boldsymbol{F}_{\rm pond} = -\frac{q^2}{4m\omega^2}\nabla|\boldsymbol{E}|^2 \propto -\frac{q^2\mathcal{A}^2}{4m}\nabla\mathcal{A}^2, \tag{54}$$

which is second order in the wave amplitude. The key feature is that the quiver speed $v_{\text{quiver}} \propto q\mathcal{A}/m$ and the cyclotron orbit speed $v_\perp \propto q\mathcal{A}/m$ both scale as $q\mathcal{A}/m$, so their ratio — the orbit radius $R = v_\perp/\omega_c \propto m/(qB) \cdot q\mathcal{A}/m = \mathcal{A}/B \propto \mathcal{A}/(k\mathcal{A}) = 1/k$ — is *amplitude-independent*, set only by the wavelength of the driving field.

The thermophoretic orbit is the static analog of this ponderomotive situation. There is no separate pre-existing transverse speed: the gradient $G_0 = |\boldsymbol{\mathcal{A}}|$ is the sole source of all motion, simultaneously providing the translational push $v_\perp \propto \varepsilon G_0$ (analog of $\boldsymbol{E}$ accelerating the charge) and the rotational bending $\Omega_3 \propto \varepsilon G_0$ (analog of $\boldsymbol{B}$ curving the trajectory). Since both responses carry the same factor εG_0, the orbit radius

$$R_{\text{orbit}} = \frac{v_\perp}{\Omega_3} = \frac{c_0\,|\Pi_\perp R_0\,\delta\mathbf{A}\,\boldsymbol{e}_3|}{|\lambda_3|} \tag{55}$$

is $O(1)$: independent of G_0 and ε, set entirely by the particle geometry through the ratio of the anisotropic mobility $\delta\mathbf{A}$ to the torque eigenvalue λ_3. This is the direct thermophoretic analog of the ponderomotive orbit radius being set by $1/k$ rather than by the wave amplitude.

When $G_0 \to 0$ the orbit does not expand (as a cyclotron orbit would if $B \to 0$ at fixed $v_\perp$): instead both $v_\perp$ and Ω_3 vanish together, and the particle freezes on a circle of fixed radius traversed infinitely slowly. There is no stored kinetic energy to preserve; the "inertia" of the orbit is provided by the particle shape, not by particle momentum.

Correspondence table. Table 2 summarises the complete mapping between the electromagnetic and thermophoretic systems.

Limitations of the analogy. Three differences are worth noting. First, $\boldsymbol{\mathcal{A}} = \nabla \ln T$ is curl-free by construction, so it cannot generate a genuine magnetic field in the electromagnetic sense; the "magnetic" character here resides in the particle's torque tensor $\mathbf{T}_{\text{th}}$, not in the field geometry. Second, the electromagnetic $\boldsymbol{E} \times \boldsymbol{B}$ drift is universal (charge-and-mass independent); the thermophoretic analog depends on particle shape through $\delta\mathbf{A}$ and λ_3, because the effective "charge" is not a scalar but a tensor embedded in the particle geometry. Third, the static thermal gradient corresponds to the *zero-frequency* limit of a ponderomotive wave ($\omega \to 0$, $k \to 0$ at fixed ω/k), so the fast quiver oscillations of the ponderomotive picture collapse to the steady spin Ω_3 itself. Despite these differences the structural mapping — a single potential field generating both a translational force and a rotational coupling through two tensor operators of different symmetry class — is exact within the linear response framework of Section 2.

3 Thermophoretic force on a surface element

We consider a surface element with area A placed in a thermal field with temperature gradient $T' > 0$ in the *negative* z-direction: $\nabla T = -T'(0,0,1) = T'(0,0,-1)$, so the gas is hotter at large z (cold below, hot above) and the thermophoretic force on a sphere is in the $+z$ direction, opposing gravity. The surface normal in spherical coordinates is $\hat{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ with tangents $\hat{t}_1 = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta)$ and $\hat{t}_2 = (-\sin\phi, \cos\phi, 0)$.

Table 2: Correspondence between electromagnetic cyclotron/ponderomotive motion and thermophoretic orbital motion of a near-sphere. Here $G_0 = |\nabla \ln T|$, ε is the asymmetry parameter, λ_3 is the dominant eigenvalue of $\delta\mathbf{T}_{\rm th}$, c_0 is the rotational drag coefficient, and $\delta\mathbf{A}$ is the anisotropic mobility tensor defined in Eq. (45).

Electromagnetic quantity	Thermophoretic analog	Expression
Vector potential $\boldsymbol{A}$	Log-gradient field $\boldsymbol{\mathcal{A}}$	$\nabla \ln T$
Time derivative ∂_t	Force tensor $\mathbf{f}_{\rm th}$	symmetric, $O(\varepsilon^0) + O(\varepsilon)$
Curl operator $\nabla\times$	Torque tensor $\mathbf{T}_{\rm th}$	asymmetric, $O(\varepsilon)$
Electric field $\boldsymbol{E} = -\partial_t \boldsymbol{A}$	Effective electric field $\boldsymbol{E}_{\rm eff}$	$-\varepsilon\, \delta\mathbf{f}_{\rm th}\, \nabla \ln T$
Magnetic field $\boldsymbol{B} = \nabla \times \boldsymbol{A}$	Effective magnetic field $\boldsymbol{B}_{\rm eff}$	$\varepsilon\, \delta\mathbf{T}_{\rm th}\, \nabla \ln T$
Charge-to-mass ratio q/m	Asymmetry-to-drag ratio	$\varepsilon\lambda_3/c_0$
Neutral particle ($q = 0$)	Sphere ($\varepsilon = 0$)	no orbit, no torque
Cyclotron frequency $\omega_c = qB/m$	Spin/orbital rate Ω_3	$G_0\varepsilon\lambda_3/c_0$
Transverse speed $v_\perp$	Horizontal drift speed	$\varepsilon G_0 \lvert\Pi_\perp R_0\, \delta\mathbf{A}\, \boldsymbol{e}_3\rvert$
Cyclotron radius $R_c = mv_\perp/(qB)$	Orbit radius $R_{\rm orbit}$	$c_0 \lvert\Pi_\perp R_0\, \delta\mathbf{A}\, \boldsymbol{e}_3\rvert/\lvert\lambda_3\rvert$
$R_c \propto 1/B$ at fixed $v_\perp$	$R_{\rm orbit} = O(1)$, indep. of G_0	εG_0 cancels in $v_\perp/\Omega_3$
Ponderomotive force $\propto \nabla\lvert\boldsymbol{E}\rvert^2$	Anisotropic thermophoretic drift	$\varepsilon G_0\, \delta\mathbf{A}\, \boldsymbol{e}_3$
Wave amplitude $\mathcal{A}$	Gradient magnitude G_0	$\lvert\nabla \ln T\rvert$

Following Eqs. (20–21) of Garcia-Ybarra & Rosner (1989), the force on the element is:

$$\frac{1}{A}\begin{pmatrix} F_{t_1} \\ F_{t_2} \\ F_n \end{pmatrix} = \alpha \begin{pmatrix} 0 \\ 0 \\ -\frac{1}{2}\sqrt{\pi/h} \end{pmatrix} - \alpha \begin{pmatrix} -\frac{a}{2}\boldsymbol{v}\cdot\hat{t}_1 \\ -\frac{a}{2}\boldsymbol{v}\cdot\hat{t}_2 \\ [2 - a(1-\pi/4)]\boldsymbol{v}\cdot\hat{n} \end{pmatrix} + \alpha\frac{3\nu}{4}\frac{T'}{T}\begin{pmatrix} \frac{a}{2}\sin\theta \\ 0 \\ (2-a)\cos\theta \end{pmatrix}, \quad (56)$$

where $\alpha = mN/\sqrt{\pi h}$. The three terms are hydrostatic pressure, viscous drag, and thermophoretic stress, respectively.

4 Thermophoretic force on a disk

A disk of area A with normal tilted by θ from $\hat{z}$ in the x-z plane has two surfaces with normals $\pm\hat{n}$. Summing both surfaces (pressure cancels, drag and thermophoresis double):

$$\begin{pmatrix} F_{t_1} \\ F_{t_2} \\ F_n \end{pmatrix} = \alpha a \begin{pmatrix} \boldsymbol{v}\cdot\hat{t}_1 \\ \boldsymbol{v}\cdot\hat{t}_2 \\ -2[2 - a(1-\pi/4)]\boldsymbol{v}\cdot\hat{n} \end{pmatrix} + \alpha\frac{3\nu}{4}\frac{T'}{T}\begin{pmatrix} a\sin\theta \\ 0 \\ 2(2-a)\cos\theta \end{pmatrix}. \quad (57)$$

5 Thermophoretic force on a cylinder

For a cylinder of radius r and length L with its axis tilted at angle θ from the X-axis in the X-Z plane, and $\boldsymbol{v} = (-v, 0, 0)$, the lab-frame force components are (Garcia-Ybarra &

Rosner 1989, Eq. 24):

$$F_X = f\left[(A\sin^2\theta + C\cos^2\theta)v + (B-D)\sin\theta\cos\theta\,\frac{T'}{T}\right], \tag{58}$$

$$F_Z = f\left[(A-C)\sin\theta\cos\theta\,v + \left(B\cos^2\theta + D\sin^2\theta\right)\frac{T'}{T}\right], \tag{59}$$

where $f = mN\pi rL/\sqrt{h}$, $A = 2(1 + a(\pi-2)/8)$, $B = \frac{3\nu}{2}(1-a/4)$, $C = a$, $D = 3\nu a/4$. For $v = 0$ and $\theta = 0$ (cylinder axis along Z), the levitation force is

$$\vec{F} = \frac{mN\pi rL}{\sqrt{h}}\,\frac{3\nu(1-a/4)}{2}\,\frac{T'}{T}\,\hat{Z}, \tag{60}$$

directed upward ($+\hat{Z}$) to oppose gravity. This is proportional to the cylinder volume $V_c = \pi r^2 L$ in the same way as the sphere result is proportional to $V_o = \frac{4}{3}\pi r^3$:

$$\frac{\vec{F}}{V_c} = \left(1 - \frac{a}{4}\right)\frac{\vec{F}_o}{V_o}. \tag{61}$$

The conversion between the Zheng (2002) and Garcia-Ybarra conventions is

$$\frac{2mN\pi\nu}{\sqrt{h}} = \frac{\eta\kappa}{\bar{v}}, \quad \bar{v} = \sqrt{2k_BT/m}, \quad \eta \overset{\text{free mol.}}{\longrightarrow} \frac{16\sqrt{\pi}}{15} \approx 1.89. \tag{62}$$

6 Hierarchy of forces and torques for a levitating particle

We now estimate the magnitudes of all four responses — thermophoretic force, thermophoretic torque, gyrophoretic force, and rotational drag torque — for a particle levitated in our thermal cell. The system parameters are: particle radius $\ell = 20\,\mu$m, particle density $\rho = 1\,\mathrm{g/cm^3}$, temperature gradient $|\nabla T| = 200\,\mathrm{K/cm}$, gas pressure $P = 5\,$Torr, and temperature $T = 300\,\mathrm{K}$ (rarefied air). The temperature gradient is directed downward ($\nabla T = -|\nabla T|\hat{Z}$), so the thermophoretic force on a sphere points upward ($+\hat{Z}$) and can balance gravity.

Gas and particle properties. The mean thermal speed of air molecules is $\bar{v} = \sqrt{2k_BT/m} \approx 415\,\mathrm{m/s}$. At $P = 5\,$Torr the mean free path is $\lambda \approx 10\,\mu$m, giving a Knudsen number

$$Kn = \lambda/\ell \approx 0.5, \tag{63}$$

placing the particle firmly in the transition regime between continuum and free-molecule flow. The particle mass is $M = \frac{4}{3}\pi\ell^3\rho \approx 3.4\times10^{-11}\,\mathrm{kg}$.

Dominant response: thermophoretic force. Using the Waldmann free-molecule result (which is a good approximation even at $Kn \approx 0.5$), the thermophoretic force is

$$F_T = \eta \frac{\kappa}{\bar{v}} \ell^2 |\nabla T| \approx 9.5 \times 10^{-10}\,\mathrm{N}, \qquad \eta = \frac{16\sqrt{\pi}}{15} \approx 1.89, \tag{64}$$

where $\kappa = 0.026\,\mathrm{W/(m\,K)}$ is the thermal conductivity of air. Gravity gives $Mg \approx 3.3 \times 10^{-10}\,\mathrm{N}$, so $F_T/Mg \approx 2.9$: the thermophoretic force is sufficient to levitate the particle, consistent with the experimental conditions.

Thermophoretic torque. The thermophoretic torque arises because different surface elements of a non-spherical body experience different stresses. Its magnitude is set by F_T acting over a moment arm of order ℓ:

$$\tau_T \sim \Lambda |\nabla \ln T| \sim \Gamma \ell |\nabla \ln T| = F_T \ell \approx 1.9 \times 10^{-14}\,\mathrm{N\,m}, \tag{65}$$

where we used $\Lambda \sim \Gamma\ell$ and $|\nabla \ln T| = |\nabla T|/T \approx 66.7\,\mathrm{m^{-1}}$. The thermophoretic torque is therefore ℓ times smaller than the thermophoretic force in the sense that $\tau_T = F_T \cdot \ell$, which is the natural geometric relationship between force and torque at the scale of the particle.

Steady-state angular velocity. A freely suspended non-spherical particle subject to τ_T will spin until the rotational drag torque $\tau_{\mathrm{drag}} = C_r \omega$ balances it. Using the free-molecule rotational drag coefficient for a sphere, $C_r = (8\pi/15)(P/\bar{v})\ell^4 \approx 4.3 \times 10^{-19}\,\mathrm{N\,m\,s}$, the steady-state angular velocity is

$$\omega_{\mathrm{ss}} = \frac{\tau_T}{C_r} \sim \frac{F_T \ell}{C_r} \approx 4.4 \times 10^4\,\mathrm{rad/s} \approx 2 \times 10^{-3} \frac{\bar{v}}{\ell}. \tag{66}$$

The steady-state spin is much slower than the natural frequency $\bar{v}/\ell \approx 2 \times 10^7\,\mathrm{rad/s}$ set by the thermal speed and particle size, confirming the slow-rotation assumption underlying the linear response theory.

Gyrophoretic force at steady-state spin. The gyrophoretic force (from angular velocity coupling) at steady-state spin ω_{ss} is

$$F_\omega \sim \Lambda' \omega_{\mathrm{ss}} = \frac{\Lambda}{\ell \bar{v}} \cdot \frac{F_T \ell}{C_r} = \frac{F_T \ell}{\bar{v} C_r} \cdot \frac{F_T}{|\nabla \ln T|} \approx 1.5 \times 10^{-9}\,\mathrm{N}. \tag{67}$$

This can be expressed analytically as

$$\frac{F_\omega}{F_T} = \frac{F_T \ell}{|\nabla \ln T|\,\bar{v}\,C_r} = \frac{\eta \kappa T}{\bar{v}} \frac{8\pi}{15\,P\,\ell} \approx 3\,Kn \approx 1.6. \tag{68}$$

The gyrophoretic force is therefore comparable to the thermophoretic force for a non-spherical particle spinning at its steady-state rate. This is perhaps surprising: although $\delta = \omega_{\mathrm{ss}}\ell/\bar{v} \approx 0.002$, the smallness of δ is compensated by the large ratio $|\mathbf{T}_{\mathrm{th}}|/|\mathbf{T}_{\mathrm{drag}}| \sim \bar{v}/\ell^2$, so that $F_\omega \sim \delta \cdot (\bar{v}/\ell) \cdot F_T/|\nabla \ln T| \cdot |\nabla \ln T| \sim F_T$. The ratio scales as Kn — it is a rarefaction effect, vanishing in the continuum limit where $\ell \gg \lambda$.

Table 3: Estimated magnitudes of the four thermophoretic and drag responses for a levitating particle of generic shape. Parameters: $\ell = 20\,\mu$m, $\rho = 1\,\mathrm{g/cm^3}$, $|\nabla T| = 200\,\mathrm{K/cm}$, $P = 5\,\mathrm{Torr}$, $T = 300\,\mathrm{K}$, $Kn \approx 0.5$. The torque column is normalised by ℓ for direct comparison with forces.

Response	Expression	Magnitude	Ratio to F_T
Thermophoretic force F_T	$\Gamma\lvert\nabla \ln T\rvert$	$9.5 \times 10^{-10}\,\mathrm{N}$	1
Gravity Mg	$\rho\frac{4}{3}\pi\ell^3 g$	$3.3 \times 10^{-10}\,\mathrm{N}$	0.35
Thermophoretic torque τ_T	$\Lambda\lvert\nabla \ln T\rvert \sim F_T\ell$	$1.9 \times 10^{-14}\,\mathrm{N\,m}^\dagger$	1
Gyrophoretic force F_ω (at ω_{ss})	$\Lambda'\omega_{\mathrm{ss}} \sim 3Kn \cdot F_T$	$1.5 \times 10^{-9}\,\mathrm{N}$	$\sim 3Kn \approx 1.6$
Rotational drag torque τ_{drag} (at ω_{ss})	$C_r\omega_{\mathrm{ss}} = \tau_T$	$1.9 \times 10^{-14}\,\mathrm{N\,m}^\dagger$	1

†Divided by $\ell = 20\,\mu$m for comparison with forces.

Summary of the hierarchy. Table 3 summarises all four responses for the present system. Several conclusions follow from this analysis.

1. **Thermophoretic force and torque are the primary drivers.** F_T levitates the particle against gravity ($F_T/Mg \approx 2.9$), while $\tau_T \sim F_T \cdot \ell$ rotates any non-spherical particle toward a preferred orientation. Both are first-order effects.

2. **The gyrophoretic force is not negligible.** For a non-spherical particle spinning at steady state, the gyrophoretic force $F_\omega \sim 3Kn \cdot F_T$ is of the same order as F_T in the transition regime ($Kn \sim 0.5$). This coupling between rotation and translation is a rarefaction effect: it vanishes in the continuum limit $Kn \to 0$. For a spherical particle the gyrophoretic coupling vanishes and this term is absent.

3. **Three small dimensionless parameters govern the dynamics.** The translational drift velocity $v \sim F_T/\Xi$ satisfies $v/\bar{v} \approx 4 \times 10^{-4}$; the steady spin satisfies $\omega_{\mathrm{ss}}\ell/\bar{v} \approx 2 \times 10^{-3}$; and the particle is small compared to the gradient scale, $\ell/L_T \approx 1.3 \times 10^{-3}$. All three confirm the validity of the linear response approximation.

4. **The transition regime requires special treatment.** With $Kn \approx 0.5$, neither the continuum ($Kn \to 0$) nor the free-molecule ($Kn \to \infty$) limit applies exactly. The full numerical solutions of the Boltzmann equation reviewed in Zheng (2002) are needed for quantitative predictions, particularly for the diagonal tensors $\mathbf{f}_{\mathrm{th}}$ and $\mathbf{T}_{\mathrm{drag}}$. The off-diagonal tensor $\mathbf{T}_{\mathrm{th}}$ has not been computed for the transition regime.